MAPP Planning Module and PR/PO: Gap Analysis

Client target (PRISM SP&A deck, SP&A Campaign workflow, context specs) compared with MAPP as built · 27 October 2026 · Scope: Planning module and the PR/PO processes attached to it

Bottom line. MAPP covers the planning hierarchy well: plans, campaigns, initiatives, activities, line items, activity approvals, All work views, the activity log and the recycle bin. The client's end-to-end SP&A workflow stops in MAPP at "activity approved". The steps after it are missing or only partly there:

- SWBS request and return
- Purchase Request (PR) and its approval
- PR number and PO number coming back from GSAP
- Invoice approval and SE/GR
- PR top-up

Some of these are not built because our constitution and decision log keep purchase requests, e-mail and live GSAP out of this phase. Building them is a **scope decision first and an engineering task second**. Inside the planning scope, about a dozen business rules work differently from what the client specifies. The biggest are: approvals tied to activities only, exchange rates, the re-approval trigger, self-approval, budget hard stop, Latest Estimate and the Zero-Based Budgeting values.

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1. Executive summary

Headline numbers

We assessed **442 items**: requirements, or tight groups of requirements, from the 16 client specs in scope, plus the 9 steps of the SP&A workflow.

Status	Items	Share	Meaning
DONE	88	20%	Built and matches the client spec
PARTIAL	110	25%	Built in part; the gap is named
MISSING	120	27%	Not built; 67 of these are purchase-request requirements
DIFFERENT	65	15%	Built, but the business rule or flow differs from the client spec
DEFERRED	58	13%	Left out on purpose by a recorded decision or the scope rules

Without the purchase-request spec (client 013), the planning side is about **24% done, 28% partial, 19% missing, 17% different and 12% deferred**. The core is solid. What remains is finishing work and aligning the rules.

How this fits with the internal build-status note ("core build done, 1–2 weeks before UAT"). The two documents measure different things, and both hold.

> - **The status note** measures MAPP against **our own specs**. It is accurate: the build is clean, all 36 test packages pass, and the open items it lists are confirmed in §8. The 1–2 weeks covers that current-scope finishing work.
> - **This report** measures MAPP against **the client's target** (deck, workflow and context specs). The PR/PO process, the planning-rule differences (§5–6) and the scope decisions (§9) are **not in that 1–2 week estimate**. They need client answers (§11) and, for PR/PO, a change of scope before they can be planned.

Ten things that matter most

#	Finding	Type	Priority
1	**Purchase requests do not exist in MAPP.** There is no PR record, form, approval, routing or top-up. Steps 2–5 and 9 of the client workflow have no support.	MISSING (scope)	High
2	**The approval engine handles Activities only.** `Round.ActivityID`, and every route is `/activities/{id}/approval`. PR, modified PR, BC, PIR, accrual, plan and budget-increase approvals cannot use it.	DIFFERENT	High
3	**The SWBS round trip is only modelled as a flag.** Line items show *Not requested / Waiting for approval / Ready to send*. Nothing is sent, no code is stored (D-255), and there are no Sent or Confirmed states.	PARTIAL	High
4	**Exchange rates differ.** The client converts with an annual planned rate per fiscal year, plus monthly rates for actuals and PRs. MAPP converts every amount with one current dated rate.	DIFFERENT	High
5	**There is no budget hard stop or budget release.** Finance budget items have no amounts, and over-allocation always warns and never blocks (D-015, D-235).	MISSING	High
6	**Latest Estimate (LE1–LE4, locked LE) is not modelled.** The client deck relies on "Planned cost (LE) vs PO vs actuals".	DEFERRED	High
7	**Business Case and PIR records are missing.** The deck has BC/PIR with approval, attachments and PPT export. Neither the client specs nor ours specify the records, and our SP&A/BCPIR views (spec 013) are Draft with no code.	MISSING	High
8	**Zero-Based Budgeting "cost hierarchy" has the wrong values.** The seed has *Media, Agency, Events, Production*; the deck defines 7 categories, from *Legally or regulatory mandated* to *Non-core spend*.	DIFFERENT	High (quick fix)
9	**Bulk actions and import (our 012) are not built.** All 11 operations are designed and in the contract but listed as not served. This also blocks "Create PR from selection".	MISSING	High
10	**The approval rules differ from the client's.** Re-approval runs on more triggers. Self-approval is off org-wide where the client allows it per workflow. An admin can decide "on behalf" where the client locks the approver. Re-Submit from Submitted is not allowed.	DIFFERENT	High

2. Scope and method

What we read:

- **The PRISM SP&A deck** (19 slides): text, tables, notes and the embedded images. The slide 11 PR flow diagram was rebuilt from its vector drawing.
- **The SP&A Campaign workflow image** (HD).
- **All 26 client specs**, with these 16 in scope:

In scope (assessed)	Out of scope (not assessed)
001 App shell · 002 Configuration · 003 People and teams · 004 Plan hierarchy · 005 Line items and cost allocation · 006 BCPIR/SP&A plans · 007 Workspace views · 008 Bulk, import, export · 009 Approval workflows · 010 Audit log and recycle bin · 011 Finance budgets (only where they feed plans) · 012 Transactions and actuals · 013 Purchase requests · 014 Accruals (only the PO-linked parts) · 015 GSAP integration · 021 Notifications (only planning and approval events) · 022 Identity and access · 026 Activity work board	016–020 Measure (reports, dashboards, attribution) · 023 AI assistant · 024 Budget optimizer · 025 Self-service dashboard builder

What we compared against:

- **Our specs:** all 29 features in `specs/`, including their open tasks.
- **Our project docs:** `docs/sdd/feature-map.md` and `decision-log.md`, the product docs (domain model, field catalogue), and the ADRs.
- **The code**, checked directly: `backend/internal/*`, `backend/api/openapi.yaml` (176 paths, 234 operations), migrations, seed, and `frontend/src/features/*`.

Five reviewers each covered one area and cited file evidence for every status. Key claims were then spot-checked in the code.

Status definitions:

- **DONE**: built and matches the client spec.
- **PARTIAL**: built in part.
- **MISSING**: not built.
- **DIFFERENT**: built, but the rule or flow differs.
- **DEFERRED**: excluded by a recorded decision (D-number) or by the scope rules in the constitution and CLAUDE.md.

> **Watch out:** the client specs use their own decision numbers (D-008, D-010, D-042 and so on), which **do not match our decision log**. For example, their D-010 is the HX key, while ours is the base currency. Agree which log is authoritative before writing new specs.

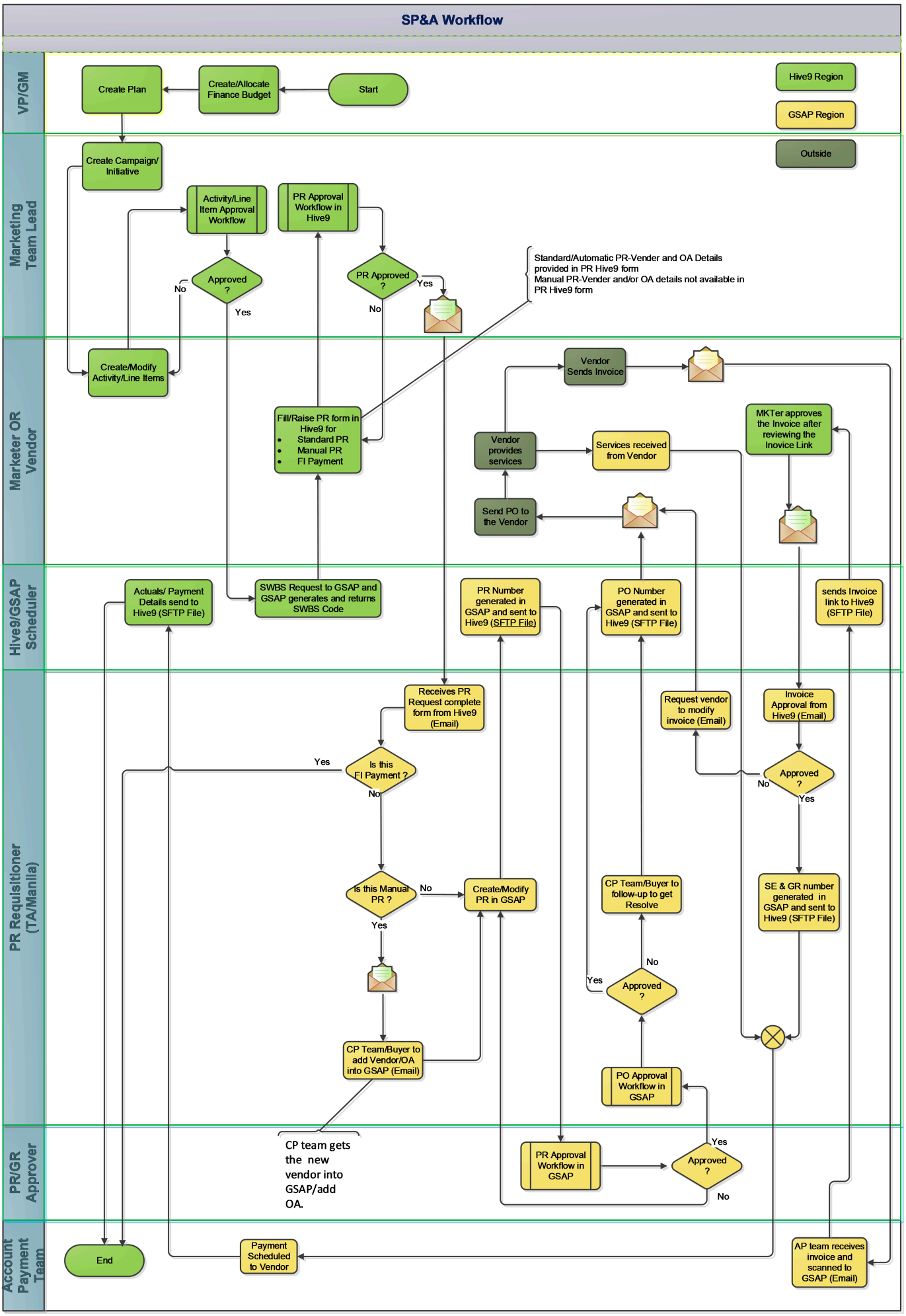
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3. The client's target process

3.1 SP&A workflow (Slide 7, seven swimlanes)

The swimlanes: VP/GM · Marketing Team Lead · Marketer or Vendor · Hive9/GSAP Scheduler · PR Requisitioner (TA/Manila) · PR/GR Approver · Account Payment Team.

Colour key: green = Hive9, yellow = GSAP, dark = outside.


SP&A Workflow
VP/GM
Marketing Team Lead
Marketer OR Vendor
Hive9/GSAP Scheduler
PR Requisitioner (TA/Manila)
PR/GR Approver
Account Payment Team
Hive9 Region
GSAP Region
Outside
Start
Create/Allocate Finance Budget
Create Plan
Create Campaign/ Initiative
Activity/Line Item Approval Workflow
Approved ?
No
Yes
PR Approval Workflow in Hive9
PR Approved ?
Yes
No
Standard/Automatic PR-Vender and OA Details provided in PR Hive9 form
Manual PR-Vender and/or OA details not available in PR Hive9 form
Create/Modify Activity/Line Items
Fill/Raise PR form in Hive9 for
Standard PR
Manual PR
FI Payment
Vendor Sends Invoice
Vendor provides services
Services received from Vendor
MKTer approves the Invoice after reviewing the Inovice Link
Send PO to the Vendor
Actuals/ Payment Details send to Hive9 (SFTP File)
SWBS Request to GSAP and GSAP generates and returns SWBS Code
PR Number generated in GSAP and sent to Hive9 (SFTP File)
PO Number generated in GSAP and sent to Hive9 (SFTP File)
sends Invoice link to Hive9 (SFTP File)
Receives PR Request complete form from Hive9 (Email)
Is this FI Payment ?
Yes
No
Request vendor to modify invoice (Email)
Invoice Approval from Hive9 (Email)
Approved ?
No
Yes
Is this Manual PR ?
No
Yes
Create/Modify PR in GSAP
CP Team/Buyer to follow-up to get Resolve
SE & GR number generated in GSAP and sent to Hive9 (SFTP File)
Yes
Approved ?
No
CP Team/Buyer to add Vendor/OA into GSAP (Email)
PO Approval Workflow in GSAP
CP team gets the new vendor into GSAP/add OA.
PR Approval Workflow in GSAP
Approved ?
Yes
No
End
Payment Scheduled to Vendor
AP team receives invoice and scanned to GSAP (Email)

#	Step (client workflow)	Lane / system	MAPP support	Evidence or gap
0	Create/allocate finance budget → Create plan → Create campaign/initiative → Create/modify activity and line items → activity/line item approval workflow	VP/GM, Team Lead, Marketer · Hive9	DONE	Global budgets set plan budgets (`internal/budget`); plan hierarchy (004–007); activity approval with levels, conditions, recall and delegation (008, 017). Line items are approved with their activity.
1	Approved → **SWBS request to GSAP**; GSAP returns the SWBS code (SFTP, 24–48 h)	Hive9/GSAP scheduler	PARTIAL	A "Request SWBS code" flag plus a derived status (`lineitem/completeness.go` `SwbsStatusOf`). No outbound file, no stored code (D-255), no Sent or Confirmed state.
2	Marketer **fills/raises the PR form** in Hive9: Standard PR, Manual PR or FI Payment	Marketer · Hive9	MISSING	No PR record anywhere (D-243; constitution scope).
3	**PR approval workflow in Hive9** (Team Lead); a decline goes back to the marketer	Team Lead · Hive9	MISSING	The approval engine is tied to activities.
4	Approved → **e-mail with the PR request form** to the PR Requisitioner. FI Payment → FI payment approval. Manual PR → CP team adds vendor/OA in GSAP	Requisitioner · GSAP	MISSING	E-mail is out of scope. MAPP would only need the PR Type and a routing record.
5	Requisitioner **creates/modifies the PR in GSAP** → GSAP **PR number** returned (SFTP) → **PR approval in GSAP**	Requisitioner, PR/GR approver · GSAP	MISSING	The feed has no GSAP PR number or PR item. The approval itself correctly happens in GSAP.
6	**PO approval in GSAP** → **PO number** returned (SFTP) → PO sent to the vendor; if not approved, the CP team follows up	GSAP, outside	PARTIAL	PO lines arrive through the simulated CSV and are matched by HX key, showing as *Committed* (015). No PO status, no PO copy link, no link to a PR.
7	Vendor delivers → invoice scanned by AP → **invoice link to Hive9** → **marketer approves or rejects the invoice** → SE/GR number returned	Vendor, AP, Marketer	PARTIAL	Invoices and credits arrive as *Invoiced*. No invoice link, no approve/reject (BUS.FIN.050 Later), no GR/SE receipt type.
8	Payment scheduled → **actuals and payment details to Hive9** (SFTP, daily)	AP · GSAP	PARTIAL	The Payment type shows as *Paid* (= actuals, D-243). Loaded by hand from a simulated file; no daily schedule (ADR-0009).
9	**Modify/top up a PR** (status *Released*): approval in Hive9 → "original PR approved in GSAP?" → "original PO generated?" → top-up approval in GSAP → modified PO to the vendor	Marketer, Requisitioner, CP	MISSING	No PR, so no modification round. "Released" is not one of the PR statuses in client spec 013, so it needs clarifying.

3.2 Purchase requisition flow stages (Slide 11)

These are the PR branches the slide shows. Each needs a decision on what MAPP does itself and what it only records.

- **Standard/Automatic PR.** Vendor and Outline Agreement (OA) details are given on the Hive9 form. The PR flows to GSAP and the PR number comes back by SFTP.
- **Manual PR.** Vendor and/or OA are not known. The requisitioner receives the form by e-mail, the CP team or buyer adds the vendor/OA in GSAP, then the PR is created in GSAP.

- **FI Payment PR.** This goes through an FI payment approval workflow, with an e-mail to CP and FM, and ends at "FI payment approved".
- **Modify/Update PR** (status *Released*). There is a new approval in Hive9. Then GSAP checks whether the original PR was approved and whether the original PO was generated. Next comes the PR top-up approval in GSAP, and the CP or buyer team modifies the PO and sends it to the vendor.
- **The legacy "3 ways to buy"** that Hive9 replaced:

    1. Check whether a Global Corporate Card (GCC) can be used.
    2. If not, contact the CP lead.
    3. A budget (WBS/GL/cost centre) is required.
    4. The PR is raised by CP SSU.
    5. MoA approval follows.
    6. The CP buyer creates the PO.
    7. GR/SE and then payment.

    No client spec covers the GCC check.

3.3 Planning calendar and features (Slides 6, 10, 13, 14)

The yearly cycle (slide 10):

1. SP&A plan kick-off and SMP planning (Jan–Mar).
2. Plan set-up and budget distribution.
3. BC development.
4. PR-PO initiation and campaign execution.
5. Actual financials and accrual financials (monthly).
6. Campaign evaluation and PIR completion (Oct–Dec).

Features the deck promises:

- **Planning:** SWBS within 24–48 h, daily GSAP actuals, approval workflows, standardised data.
- **PR/PO:** PR/PO raised in Hive9, PR/PO number back in 24–48 h, PO budget limits and thresholds, attachments.
- **Budget control:** budget hard stop, Latest Estimate vs PO (commitment) vs actuals.
- **BC/PIR:** BC linked to activities with Marketing and Finance approval, PIR with PPT export.

ZBB cost hierarchy (slide 14): Legally or regulatory mandated costs · Long term contracts · Core business defend & grow · Strategic priorities · Growth drivers · Tactical investments · Non-core spend. "If in doubt, the Marketing GM makes the final call."

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4. Coverage by area

Area	Client specs	Unit	DONE	PARTIAL	MISSING	DIFFERENT	DEFERRED
A. App shell, configuration, people, access	001, 002, 003, 022	requirement groups	22	25	12	16	4
B. Plan hierarchy, line items, BC/PIR & SP&A	004, 005, 006	requirement groups	20	13	10	16	7
C. Views, bulk/import, audit log, work board	007, 008, 010, 026	requirement groups	17	14	20	12	10

Area	Client specs	Unit	DONE	PARTIAL	MISSING	DIFFERENT	DEFERRED
D. Approvals and notifications	009, 021 (approval part)	requirement groups	20	16	11	7	4
E. PR/PO, actuals, GSAP, budgets	013, 012, 015, 011/014 parts	single FRs	9	42	67	14	33
Total	16 specs		**88**	**110**	**120**	**65**	**58**

5. Findings by area

A. App shell, configuration, people and access (client 001, 002, 003, 022)

Works well:

- one layout, breadcrumbs, bookmarkable addresses, all errors at once, the unsaved-changes guard, version conflicts, delete with Undo, help tips;
- teams, the person picker and approval delegation;
- server-side access that fails closed, with scopes and 404 outside them, and a full audit of access changes.

Req	Requirement (short)	Status	Gap / evidence	Priority
001 FR-015–017; 002 FR-059–063, FR-078	Convert with the fiscal-year annual planned rate (monthly for actuals and PRs), show the rate basis, copy rates from the previous year	DIFFERENT	One dated "current" rate converts everything (016 FR-002, `settings/rates_book.go`); units per USD (D-010).	High
003 FR-016/017/022/023	"Can Edit Team Content?" and "Can Approve?" on each team membership	DIFFERENT	No flags on `people.Membership`; roles and scopes do this job (D-223).	High
003 FR-021	Owners at every level must belong to the plan's teams	PARTIAL	Checked for the plan owner only. Activity, campaign and initiative owners can be any active person (`activity/fields.go` `checkOwner`).	High
022 FR-030–033	Only the assigned approver or their delegate decides; self-approval set per workflow	DIFFERENT	`approvals.decideOnBehalf` lets the Administrator decide; self-approval is one global rule (D-302).	High
022 FR-019–021	Screens hide or disable forbidden actions with the reason; NoAccess on 403	PARTIAL	The server is done. Screens are not: 014 tasks T043–T051 are open.	High
002 FR-008–011	Shared master-data lists; values with code, label and description ("code - label")	DIFFERENT	Options carry a label only and belong to one attribute; GL and cost-centre codes are missing.	High
002 FR-012–014	List import: check everything, import nothing on any error, update by code, 20k rows	DIFFERENT	Adds new valid rows only, skips existing ones, up to 5,000 rows.	Med
002 FR-016/017, FR-019	Attribute types date, currency, decimal, attachment; min/max; "required to save"	PARTIAL	No such types; markings are optional, needed to publish, or needed to submit.	Med
002 FR-020–024	Calculated, roll-up and lookup attributes; approvable attribute groups	MISSING	—	Med

Req	Requirement (short)	Status	Gap / evidence	Priority
002 FR-029–035	Dependent options from an ancestor record; visibility conditions beyond Business Unit	PARTIAL	Same-level parent only; Business-Unit-only conditions.	Med
002 FR-065–070, FR-080	Organisation settings (plan budgets, AOP, fit dates, require finance-budget link, time-out…)	MISSING	Only the access rules exist.	Med
002 FR-077	Per-model "Re-approve when planned cost changes" switch	DIFFERENT	A fixed rule for every model (`activity/reapproval.go`).	Med
002 FR-005/006; 003 FR-008	In-use configuration and people deleted to the bin with a "(deleted)" marker	DIFFERENT	Set inactive instead; people are never deleted.	Med
022 FR-010–016	One role per person from a Shell group; 20 Hive9 roles	DIFFERENT	Five editable roles, several per person, with team scopes (D-223, Q-010).	Med
022 FR-024	Financials-only visibility shows money as "Restricted"	DIFFERENT	`finance.view` hides only GSAP figures.	Med
003 FR-011–013	Import users from a spreadsheet	MISSING	—	Med
001 FR-012–014	Display-currency switch in the header	PARTIAL	Only in My profile.	Med
022 FR-002–009	SSO with MFA, time-out warning	DEFERRED	D-004, ADR-0005: demo sign-in until SSO.	—

B. Plan hierarchy, line items, BC/PIR and SP&A (client 004, 005, 006)

Works well:

- the hierarchy and its validation; budget flowing down and cost rolling up; remaining budget that may go negative;
- model goals and the waterfall;
- line-item spreads with the unassigned balance (Sys_Gen_Balance);
- copy rules (no approvals or GSAP values); activity statuses; audit.

Req	Requirement (short)	Status	Gap / evidence	Priority
005 FR-014/015	SWBS code read-only from GSAP; GSAP status Draft/Approved/Sent/Confirmed; lock "Create SWBS" once sent	PARTIAL	Derived status only; no stored code (D-255); nothing sent.	High
005 FR-041, FR-048	"Purchase Requests & POs" section and Create PR; planned cost not below the PR amount	MISSING	D-243: no purchase requests. POs are read-only (015).	High
005 FR-039	Hard limit / remaining-budget check when linking (hard stop)	MISSING	Finance budget items have no amounts (`settings/model.go`).	High
004 FR-056	Latest Estimate LE1–LE4 and Locked LE per round and year	DEFERRED	Feature-map "Later"; no fields or API.	High
006 FR-004–021	SP&A view: planned, committed, actual, remaining, utilisation, variance by quarter and channel	MISSING	Our 013 is Draft (gated on Q-001); no code.	High
Deck	Business Case record linked to activities, with approval and attachments; PIR with approval and PPT export	MISSING	Only a "PIR" yes/no attribute (`see d/settings.go:148`); no attachments anywhere.	High

Req	Requirement (short)	Status	Gap / evidence	Priority
004 FR-036	(Zero-Based) Cost Hierarchy with the 7 ZBB values; Lubricants-only	DIFFERENT	Seeded as "Media, Agency, Events, Production" for all BUs (`seed/settings.go:153`).	High
004 FR-034	Activity dates may cross fiscal years and sit outside the initiative	DIFFERENT	R3 requires the activity to fit inside its initiative (`activity/dates.go`).	Med
005 FR-002	Line-item changes never change the activity's status	DIFFERENT	Line-item add, delete, cost or currency restarts approval (D-013).	Med
005 FR-016–018	Convert at the annual planned rate of the plan's fiscal year; refuse when it is missing	DIFFERENT	Latest dated rate (D-258); plan currency (D-260).	Med
004 FR-066; 005 FR-044	Refuse delete while transactions or PRs are linked	DIFFERENT	Allowed with a warning (015 FR-029).	Med
004 FR-017	Refuse a model change when activities use a disallowed type	DIFFERENT	Allowed with a warning; those activities are flagged.	Med
004 FR-012	Warn and confirm when the creator is in none of the plan's teams	DIFFERENT	Refused instead; visibility follows role scope (D-226).	Med
004 FR-024/025	Campaign group approval (GM sign-off); campaign Submit cascade	DEFERRED	D-012: derived approval progress only.	Med
004 FR-037	Separate cost and performance splits (each 100%)	PARTIAL	One share per option.	Med
004 FR-038; 005 FR-019	Total PR amount, % of actual cost, Actuals @ monthly FX	PARTIAL	Committed, Invoiced, Paid and Accrued from the simulated feed; no PR amount or monthly FX.	Med
005 FR-036/037	Finance budget link per budget year, with the remaining amount in the picker	PARTIAL	Items have no amounts, so there is no "remaining".	Med
004 FR-052–054	Multi-year activities, "Link to" per year	DEFERRED	D-017.	Med
006 FR-001–003	BCPIR marker and filter wherever plans are listed	PARTIAL	The flag is stored and logged; no marker or filter.	Med
004 FR-009	Plan year from now to +10	DIFFERENT	We allow −1 to +5.	Med
004 FR-013 etc.	Organisation toggles (AOP, budget configuration, shrink dates, unallocated…)	MISSING	No organisation-settings record.	Low

C. Views, bulk and import, audit log, work board (client 007, 008, 010, 026)

Works well:

- the three All work views (grid, budget & cost, calendar); roll-ups; column chooser; personal saved views; AOP and Show levels; export to .xlsx and .csv;
- the activity log on every record; write-once audit;
- the recycle bin with restore-with-parents and purge.

Req	Requirement (short)	Status	Gap / evidence	Priority
008 FR-001–010, 013–016, 019, 022–025	Bulk select, change owner, change field, copy, move, delete with per-item results	MISSING	Designed in our 012 (D-126–D-139); 11 contract operations in `notYetServed` ; no screens.	High

Req	Requirement (short)	Status	Gap / evidence	Priority
008 FR-044–051	Import activities from a template, validated, with undo	MISSING	Designed in 012; row limit 1,000 (ours) vs 5,000 (client) is still open.	High
007 FR-004, FR-020, FR-022–028	Row actions and inline editing in the workspace	DEFERRED	All work is read-only on purpose (010 FR-002) so approval restarts are never silent.	High
010 FR-002/004	Audit fields: source (User/Import/GSAP/System/Restore), on-behalf, year, operation reference	PARTIAL	`activity_log` has none of these.	High
007 FR-043/044	Drag on the calendar to reschedule	DIFFERENT	Not draggable on purpose (010 FR-027).	Med
007 FR-071, FR-085	Filters, view and options remembered per user across sessions and devices	DIFFERENT	Only columns and sort are stored; filters last for the session (D-122).	Med
007 FR-079–082	Share saved views with teams or users	MISSING	Views are personal only (010 FR-030).	Med
007 FR-034–038	8 summary cards with Annual/Total, Allocated, Actual @ FX; hide/reorder	PARTIAL	6 fixed cards.	Med
007 FR-047/048	Budget & Cost per period including Actuals	PARTIAL	No actual cost per period.	Med
007 FR-052–062	Attribute filter panel per level, line-item filters, date sets	PARTIAL	The backend exists; the UI (016 T074–T077) is open.	Med
010 FR-011	Separate Plan, Finance and Configuration logs	DIFFERENT	One log with a type filter.	Med
010 FR-018	Export the audit log	MISSING	—	Med
010 FR-043–045	Retention (bin 90 days, audit 7 years) with a daily job	DIFFERENT	Nothing is purged automatically; the write-once trigger would block audit retention.	Med
008 FR-030/031	Copy Plan 1–10 copies, named `_Copy_MMDDYYYY_HHMMSS_n`	DIFFERENT	One copy with a "(copy)" suggestion.	Med
008 FR-017	Create Purchase Request from a selection	DEFERRED	Purchase requests are out of scope.	—
026 all	Activity Work Board (kanban, columns, WIP, swimlanes)	MISSING	Nothing exists. The client marks it "Proposed, needs Shell approval".	Low
008 FR-018, FR-037–043, FR-057–067	Templates, snapshots, calendar feed	DEFERRED	Feature-map "Later".	Low

D. Approvals and notifications (client 009, 021)

Works well:

- serial levels with Any or All; amount and attribute conditions; frozen rounds; auto-approval; recall; the stale-action conflict (409);
- delegation, reminders, escalation, Monitoring; a complete approval history; the inbox;
- re-approval warnings; in-app notifications for approval events.

Req	Requirement (short)	Status	Gap / evidence	Priority
009 FR-004, FR-016, FR-034/035	Workflow object types: Activity, **PR, Modified PR, Accrual**, attribute groups	PARTIAL	Activity only (`approval/model_rounds.go` `ActivityID`).	High
009 FR-003	Per-workflow settings: threshold currency, comment on decline/approve, allow self-approval (default **yes**)	DIFFERENT	None on `Workflow` ; self-approval is globally "never" (D-280, D-302).	High
009 FR-008	Conditions on Country/OU and custom attributes	PARTIAL	Planned cost (USD), business unit and activity type only.	High
009 FR-050	Re-approval on a **planned-cost** change, per model setting	DIFFERENT	Always on; also triggers on dates and line-item add, delete, amount or currency (D-013).	High
009 FR-014	Bulk-load 100+ workflow definitions from a file	MISSING	—	High
009 FR-017	Re-Submit from Submitted, cancelling the open round	DIFFERENT	Created or Declined only (`rounds_submission.go:80`); Submitted needs a Recall first.	Med
009 FR-027	Self-exclusion empties a level → refuse the submission	DIFFERENT	Falls back to the backup approver or an on-behalf decision.	Med
009 FR-030	In-Progress and Complete set by a status change	DIFFERENT	Derived from dates.	Med
009 FR-041–044	Approvals queue: item type, dates, status columns, per-row actions, sort, paging, saved views	PARTIAL	Activities only; actions are in the preview panel.	Med
009 FR-029	Bulk submit	MISSING	Depends on 012.	Med
009 FR-048/049	Gates: SWBS (done), PR submission, external-name generation	PARTIAL	No PR gate; the external name is free text.	Med
021 FR-001–005	Per-user notification rules (event × channel), org defaults	MISSING	Fixed in-app approval notifications only.	Med
021 FR-020, FR-023–029	Notification List page; custom and due-date notifications	MISSING	—	Med
021 N01–N14	Planning events (edited, deleted, assigned, due date…)	MISSING	Notifications live inside `approval` .	Med
009 FR-033	Plan Draft → Published with **no Unpublish**	DIFFERENT	Unpublish exists (`plan/status.go:61`).	Low
009 FR-025; 021 N04/05	Notify the owner **and the submitter** of a decision	PARTIAL	Owner only (`rounds_decide.go:207`).	Low
021 e-mail	E-mail channel	DEFERRED	E-mail is out of scope (constitution).	—

E. Purchase requests, transactions and actuals, GSAP, budgets (client 013, 012, 015, 011/014 parts)

Works well:

- a read-only, simulated GSAP feed of PO lines, invoices (credits negative), payments and accruals;
- check-then-import, idempotent re-runs, matching by line-item ID → HX key → PO → SWBS, manual linking of unmatched rows;

- Planned, Committed, Not yet committed, Invoiced and Paid at every level; a Financials page; finance permissions.

Req	Requirement (short)	Status	Gap / evidence	Priority
013 FR-001–036, 048–071	**The whole PR lifecycle**: create from a line item, header and lines, one company code, balance check, reservation, submit gates, approval and modification workflows, close, delete or restore, list page, audit	MISSING	No PR concept (D-243; constitution scope). The reusable parts already exist: `record.Meta`, `history`, the approval engine (once it handles any record type), Shell attribute levels.	High
013 FR-023–025	PR reservation from the first save; release rules; roll-up to the activity	MISSING	Nothing reserves money; Committed comes only from GSAP PO lines.	High
013 FR-026	Refuse deleting a line item or activity that a PR uses; refuse lowering planned cost below the PR total	DIFFERENT	Delete is allowed with a warning (015 FR-029).	High
013 FR-030	Submission gates: activity approved, SWBS code present, required fields filled	MISSING	Line items store no SWBS code (D-255).	High
013 FR-039–045	Route the approved PR form to the requisitioner (Excel), with retries and routing history	DEFERRED	E-mail is out of scope. Recommended: in-app routing plus a downloadable PR form.	High
015 FR-006–016	Daily FSSI 1523 SWBS request file and SWBS code return	DEFERRED	"Sending to GSAP (future)"; ADR-0009 covers inbound feeds only.	High
015 FR-017/019	Inbound fields: GSAP PR number and item, PO status, PO copy link, invoice status and link, GR/SE receipts	PARTIAL	The CSV has type, document, PO, date, amount, currency, vendor, SWBS and HX only.	High
015 FR-018	Matching M1–M8: attach to an approved PR line; company-code check	DIFFERENT	D-245 matches without PRs and without a company-code check.	High
015 FR-035–037	Finance exception list: Reprocess, Assign, Dismiss, last seen	PARTIAL	A "Not linked" tab plus manual link; no Dismiss or exception records.	High
015 FR-038/040	GSAP status Draft/Approved/Sent/Confirmed on the line item, with PR and PO data	PARTIAL	Three derived statuses only.	High
015 FR-041–043	Company-code rule	MISSING	Company code exists only as an activity attribute.	High
013 FR-017; 012 FR-016/017	Balance check at the monthly PR rate or the annual planned rate; actuals at the annual rate plus @ monthly FX	DIFFERENT	One current rate (D-303).	Med
013 FR-046/047	Supporting documents (10 MB per file, 20 MB per PR)	DEFERRED	BUS.FIN.060 Later; no upload capability in MAPP.	Med
013 FR-058–060	Marketer approves or rejects vendor invoices	DEFERRED	BUS.FIN.050 Later.	Med
012 FR-009–014	Split one transaction across line items; Attributed and Remaining	DIFFERENT	One whole link per transaction (D-253).	Med
015 FR-020/021	Latest business date wins; cancelled invoices leave the totals	PARTIAL	The last import wins; no invoice status.	Med
015 FR-034	Archive every file with its outcome	MISSING	The file is read in memory only.	Med

Req	Requirement (short)	Status	Gap / evidence	Priority
011 FR-024/025	Hard limit: refuse a change that would push the remaining budget below zero	DIFFERENT	Warn only (D-015, D-235; BRD BUS.BUD.030 "warning, not a blocker").	Med
013 FR-061–064	Finance › Purchase requests page grouped Vendor › PR › line › line item	MISSING	The Financials page lists transactions only.	Med
Deck	"3 ways to buy": GCC check before a PR	MISSING	Not in any client spec; needs clarifying.	Low

6. Business logic and flow changes needed (consolidated)

Ordered by dependency. Each item names the packages it touches. Sizes are relative: S, M, L, XL.

#	Change	Why (client source)	Touches	Size
L1	**Approve any record type.** Make the round's subject generic: `Round.ActivityID` becomes `SubjectKind` + `SubjectID`, and `Workflow.ObjectType` is added. Add one facts adapter per type, generic decide routes and an object-type filter on the inbox.	009 FR-004/016/034; deck "multi-layered approval for SP&A Plan, BC, PIR, Accrual, PR"	`approval/*`, `app/approvals_adapter.go`, `access/routes_*.go`, openapi, `features/approvals`	L
L2	**Per-workflow settings:** comment on decline/approve, allow self-approval, threshold currency. Agree how they relate to the global segregation-of-duties rule (D-280, D-302) and to the admin "decide on behalf" right.	009 FR-003, FR-027; 022 FR-030–033	`approval/model.go`, `rounds_decide.go`, `resolve.go`, `access/rules.go`	M
L3	**Re-approval rule.** Agree the triggers: client = planned cost only, switched per model; ours = cost, dates, line items (D-013). Make it a model setting.	009 FR-050; 002 FR-077; 005 FR-002	`activity/reapproval.go`, `lineitem/create,update,delete.go`, `settings` models	M
L4	**Store SWBS on the line item.** Add `swbsCode` plus a GSAP status Not requested → Waiting → Sent → Confirmed, with sent and confirmed dates and a 24–48 h SLA. Add a simulated outbound SWBS request file and a code upload (check-then-import). Reverses D-255; needs a new ADR for outbound feeds.	Deck step 1; 005 FR-014/015; 015 FR-005–016	`lineitem/*`, migration, `financials` matching, `approval/hooks.go`	L
L5	**Purchase Request record** (new `purchaserequest` package; see the model in §7). Hangs off line items; reserves money; has its own and modification approval; GSAP fields come only from the feed.	Deck steps 2–5, 9; 013	new package, `lineitem`, `activity`, `financials`, `approval`, `access`, `settings` (PR attribute level), new screens	XL
L6	**GSAP matching to PR lines and richer inbound fields.** PR number and item, PO status and copy link, GR/SE receipts, invoice status and link, company-code check (M5–M7), supersede by business date, exception records with Dismiss.	015 FR-017–021, 035–037, 041–043	`financials/*`, CSV layout (D-247), migrations	L
L7	**Exchange rates by fiscal year.** Annual planned rate per currency and year for planned and budget amounts; monthly rates for actuals and PRs; copy year to year. Reopens D-010, D-027, D-258, D-303.	001 FR-015–017; 002 FR-059–063; 005 FR-016–018; 012 FR-016/017	`settings/rates*`, `money`, `lineitem`, `rollup`, `allwork`, `budget`, `financials`	L

#	Change	Why (client source)	Touches	Size
L8	**Budget hard stop and release.** Amounts and a hard-limit flag on finance budget items (or on global budgets); remaining-budget check on line-item save or link; release when a PR closes; specify the empty `019-budget-increase-approval` as the increase/release approval.	Deck "Budget release & hard stop"; 005 FR-039; 011 FR-024/025	`settings` finance items, `budget`, `lineitem/validate.go`, L1	M–L
L9	**Latest Estimate.** LE1–LE4 plus locked copies on the activity; an LE-lock record per round and year; LE column in All work; LE vs PO vs actuals.	Deck "Planned cost (LE) vs PO vs Cost"; 004 FR-056	`activity`, `allwork`, `financials`, migration, UI	M
L10	**BC and PIR records.** BC linked to one or more activities, with approval (L1) and attachments; PIR as a dated review with approval and PPT export. Answer Q-001 first, then build our 013 SP&A and BCPIR views.	Deck slides 6, 13, 16; 006	new package(s), `plan`, `approval`, file storage (ADR)	L
L11	**Owners from the plan's teams at every level**, plus membership flags "Can Approve?" and "Can Edit Team Content?" if the client keeps them.	003 FR-016–023	`activity/fields.go`, `campaign/*`, `people/*`, `approval/resolve.go`	M
L12	**Delete protection.** Refuse deleting a line item, activity, campaign or plan with linked transactions or PRs (client) instead of warning (015 FR-029).	004 FR-066; 005 FR-044; 013 FR-026	`lineitem`, `activity`, `campaign`, `plan` delete paths	S
L13	**Activity dates.** Relax R3 (activity inside its initiative) if cross-year activities are needed; this ties into multi-year linking (D-017).	004 FR-034, FR-052–054	`activity/dates.go`, `rollup`	M
L14	**Master data with codes.** Option code and description, or a shared List entity; atomic import that updates by code.	002 FR-008–014	`settings/*`	M
L15	**Audit model.** Add source, on-behalf, year, plan and operation ID; export; retention with a controlled exception to the write-once trigger (ADR).	010 FR-002/004, 018, 043–045	`history`, migration, `activity-log` UI	M
L16	**Notifications as a module.** Move them out of `approval`; add an event catalogue (planning events, PR events), per-user rules, a list page, notify the submitter as well as the owner.	021	new `notification` package, UI	M
Q1	**Quick fixes:** ZBB cost hierarchy values (`seed/settings.go:153`); Re-Submit from Submitted; plan year range to +10; BCPIR marker and filter; Unpublish (keep or remove); display-currency switch in the header.	004 FR-036, FR-009; 009 FR-017, FR-033; 006 FR-001–003	seed, `approval`, `plan`, `TopBar.tsx`	S each

7. Proposed Purchase Request model (for discussion)

PurchaseRequest (`PR-1001`, embeds `record.Meta`):

Field	Notes
Basics	name, description, owner, creator
`planId`	for scope; all lines in one plan
`companyCode`	from the first line's activity; one per PR
`type`	Standard PR/PO · Manual PR/PO · FI Payment

Field	Notes
`status`	Created → Submitted → In review → Approved / Declined → Closed. **"Released" needs mapping.**
`approvalWorkflowId` , `modificationWorkflowId`	
Attributes	Material/Service, short text, vendor, vendor code, Outline Agreement, MoA approver, e-mail group, recipients; China-only OA vendor and sub-category
`gsapPrNumber` , `gsapPrDate`	read-only; filled by the feed
`approvedSnapshot`	used during a modification round

PurchaseRequestLine:

Field	Notes
`lineItemId`	unique per PR
`hxKey`	"HX" + line-item number
Money	currency, amount (minor units), overall limit, delivery date, closed amount
PO fields	PO number, line, status, date, value, copy link; feed only

Rules:

- Reserved = the sum of live PR lines, with a Closed PR counted at its closed amount. It is checked in the line item's currency, inside the same `Write` transaction.
- Planned cost cannot drop below the reserved amount.
- A PR can be submitted only when the activity is Approved and the SWBS code has been received.
- **Routing without e-mail:** an in-app routing record plus a downloadable PR form (Excel or CSV, with the requisitioner's columns), and optionally a requisitioner queue for the TA/Manila role.

Access: new permissions `pr.view` , `pr.edit` , `pr.submit` , `pr.route` , `pr.decideInvoice` , with route rules; a Requisitioner role.

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8. Work still open on our side (inside the current scope)

These are open in our own feature map and task lists, whatever the client specs say. Each was confirmed as an unticked task in `specs/*/tasks.md` . The last column shows whether the internal build-status note lists it.

Feature	Open items	In status note?
012 Bulk actions and import	Spec **Draft**. The contract and access rules exist, but the 11 operations are listed as not served in `contract_routes_test.go` and answer 404; there are no screens. **Decision needed: build it, or take it out of the contract.**	Yes
016 Planning and currency	Attribute filters, Breakdown tab and "Planned cost by" card (T074–T077); field-help tips on the record screens (T084–T088)	Yes
016 Planning and currency	Import GSAP rates route (T028); e2e (T089)	No

Feature	Open items	In status note?
014 Access control	Role-aware screens (T043–T051): a Viewer still sees buttons they can't use. The API refuses the action, so this is a **UX gap, not a security hole**.	Yes
014 Access control	e2e for the access journeys (T061)	No
017 Approval rules	Approvals read roles from a stand-in rather than from 014 (T001, T008); line-manager field on People (T036)	Yes
017 Approval rules	e2e (T047–T049)	No
015 Financial transactions	Linked-transactions sentence in some delete dialogs (T045)	Yes
015 Financial transactions	e2e (T021, T025, T030, T035); Home no-access state (T044)	No
008 Activity approvals	e2e (T055–T056)	No
001 App shell	End-to-end accessibility test (T080)	No
013 BCPIR and SP&A	Spec **Draft**, gated on Q-001; no code	No
019 Budget-increase approval	**The spec folder is empty**; there is no spec and no code	No

Missing from the status note: the end-to-end tests (12 tasks across 001, 008, 014, 015, 016 and 017), the 013 and 019 specs, and two small items (016 T028, 015 T044). The e2e gaps matter before UAT: CLAUDE.md requires an end-to-end test for every P1 story.

8.1 Build health, checked against the status note

Status-note claim	Measured on 27 Oct 2026	Verdict
Backend compiles clean	`go build ./...` succeeds	Confirmed
36 of 36 backend test packages pass	`go test -short ./...` : 36 ok, 0 failed (22 packages have no tests)	Confirmed
1,501 Go tests	1,501 `func Test…`	Confirmed
Zero TODO/FIXME in Go	0	Confirmed
~78,000 lines of Go	~74,300 lines of non-test, non-generated Go; ~143,500 including tests and sqlc output	Depends on how you count
665 frontend source files, 351 test files	728 non-test `.ts` / `.tsx` files (excluding generated API types); 351 test files	Test files confirmed; source count depends on how you count
41 browser tests	40 `test(...)` cases in 15 Playwright files	Off by one
176 API endpoints	176 paths, 234 operations	Confirmed (as paths)
Every sidebar section has a real page	`COMING_SOON_SECTIONS` is empty and no Settings card is marked "coming soon" (the mechanism is still in the code)	Confirmed

Status-note claim	Measured on 27 Oct 2026	Verdict
GSAP transactions, rates, Entra ID sign-in and the nightly approvals job are simulated by design	Matches ADR-0005, ADR-0009, D-004, D-242, D-259, D-308	Confirmed; the client-target gaps for these are in §5 E and §9

The short test run skips the scope-isolation tests that need PostgreSQL (`-short`). The full `go test ./...` run with `TEST_DATABASE_URL` was not part of this check.

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9. Scope decisions needed before PR/PO work starts

1. **Bring purchase requests into scope.** This needs a constitution amendment (Current Phase & Scope), CLAUDE.md non-negotiable 13, feature-map rule 6, and replacing D-243. The BRD marks BUS.FIN.020 as Future/Later.
2. **E-mail.** Keep it excluded and route PRs in-app with a downloadable PR form, or allow outbound e-mail through an adapter (constitution change).
3. **Outbound files to GSAP** (SWBS request, PR data). Needs a new ADR, since ADR-0009 covers inbound simulated feeds only, and reversing D-255.
4. **BRD "Later" items moving forward?** Invoice approval (BUS.FIN.050), documents and PO PDFs (BUS.FIN.060), budget hard stop (BUS.BUD.030, which conflicts with D-015 and D-235).
5. **Finance budgets module and accrual calculation** stay out unless decided otherwise. PR reservation does not need them.
6. **The actuals model.** Keep Paid = actuals at the current rate (D-243, D-248, D-303), or move to CO postings, split links and annual plus monthly FX (client 012).
7. **Which decision log is authoritative.** The client specs' D-numbers clash with ours.

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10. Recommended roadmap

Phase	Content	Depends on
0. Finish for UAT (the status note's 1–2 weeks)	Every open task in §8, including the missing e2e tests · decide whether 012 is built or taken out of the contract · Q1 quick fixes · write or remove the empty 019 spec	—
1. Planning rule alignment	L2 per-workflow approval settings · L3 re-approval rule · L11 owners and team flags · L12 delete protection · L14 master-data codes · build 012 bulk and import	Client answers Q2–Q8
2. Engine and finance foundations	L1 approve any record type · L7 fiscal-year rates · L9 Latest Estimate · L8 hard stop and release · L15 audit fields	Phase 1; rate decision
3. PR/PO (after the scope decision)	L4 SWBS round trip · L5 Purchase Request · L6 GSAP matching to PR lines · PR list page · in-app routing and PR form	§9 decisions 1–3
4. BC/PIR and SP&A	L10 BC and PIR records, SP&A and BCPIR views, PPT export · L16 notifications module	Q-001; L1
Later	Invoice approval, documents, Work board (026), templates, snapshots, calendar feed, multi-year linking	Client priorities

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11. Open questions for the client

Purchase requests and GSAP

1. Is a PR raised in MAPP and sent to GSAP (as in Hive9), or raised in GSAP and only shown in MAPP? Can one PR span several line items or activities?
2. What does **"Released"** mean (Modify/Update PR in the deck)? It is not one of the client 013 PR statuses.
3. **PR types:** do FI Payment and Manual PR change MAPP's behaviour (for example, no PO data, vendor not required), or only the requisitioner's?
4. Does a PR reservation count as committed spend (013 FR-023) or sit inside planned cost (011)? Should "Committed" stay PO-only?
5. Without e-mail, is in-app routing plus a downloadable PR form acceptable for the requisitioner (TA/Manila) and AP?
6. **SWBS:** one code per line item or per activity? Is it requested automatically on final approval, or only when "Create SWBS" is ticked? What if GSAP rejects the request or misses the 48-hour window?
7. Which GSAP extract layouts (FSSI 1523, PGSI 941) and file names apply during the freeze?
8. Should MAPP guide the "3 ways to buy" choice (GCC check) before a PR?

Planning rules 9. **Exchange rates:** a fixed annual planned rate per fiscal year, or the latest dated rate? Should a rate change alter earlier years' USD values? 10. **Re-approval:** planned cost only, switched per model, or our broader triggers? 11. **Self-approval and on-behalf:** set per workflow with "allowed" as the default? May an Administrator decide on behalf of someone else? 12. **Hard stop:** at which level (finance budget row, plan, global budget)? Block or warn? Who approves a budget increase or release? 13. **Latest Estimate:** who locks a round, and when? Per activity or per line item? Phased by month? In USD or the plan currency? 14. **BC/PIR:** is the Business Case a record of its own with attachments and approval? What goes into the PIR PPT export? 15. **ZBB cost hierarchy:** confirm the 7 values, single or multiple choice, the level, and whether it is limited to Lubricants. 16. **Multi-year:** must activities be allowed to cross fiscal years and sit outside their initiative's dates? 17. **Roles:** map VP/GM, Team Lead, Marketer and Requisitioner onto our five roles plus scopes, or seed Hive9's 20 roles? 18. **Views and editing:** are inline editing and drag-to-reschedule must-haves if every edit shows "this sends N activities back for approval"? 19. **Deletes and retention:** refuse deletes with transactions or PRs? What are Shell's retention periods for the bin and the audit log? 20. **Work board (026):** has Shell approved it for this phase?

Appendix: evidence

Client material:

- `context/PRISM SP&A Demo v1.0 1 (1).pptx` : slides 4–7, 9–11, 13, 14, 16, 18.
- `context/Slide 7 – SP&A Campaign workflow (HD).png` .
- `context/specs/*` : 16 specs in scope.

MAPP sources:

- `specs/001–029` and `docs/sdd/feature-map.md` , `decision-log.md` .
- `docs/product/domain-model.md` , `field-catalogue.md` .
- `backend/internal/{plan,campaign,activity,lineitem,approval,financials,budget,settings,people,access,history,allwork,rollup}` .
- `backend/api/openapi.yaml` (176 paths, 234 operations) and `backend/seed/settings.go` .
- The internal build-status note (§8.1), checked on 27 October 2026 with `go build ./...` , `go test -short ./...` , counts over the source tree, `contract_routes_test.go` and the open tasks in `specs/*/tasks.md` .
- `frontend/src/features/*` .

Spot-checked in the code:

- ZBB seed values (`seed/settings.go:153–154`).
- Plan Unpublish (`plan/status.go:61`).
- Re-submit only from Created or Declined (`approval/rounds_submission.go:80`).